

Executive Reporting Agent (Agent 6)

Deep Technical Specification & Operational Manual

Executive Overview & Purpose

The Executive Reporting Agent is the final synthesis node in the CORE pipeline graph. It aggregates all sub-agent findings into executive summaries, computes session risk ratings, and powers multi-format export engines for PDF, Markdown, and JSON reports.

1. Architectural Overview & Workflow

- Position in LangGraph Engine: Mandatory final node across ALL pipelines (Pipeline A, B, C, A_THEN_B, CUSTOM).
- State Synthesis: Consolidates `state.triage_report`, `state.remediation_plan`, `state.code_audit_report`, `state.compliance_report`, and `state.threat_intel_report`.
- State Propagation: Writes `state.executive_summary` and triggers backend report generation.

2. Multi-Format Report Generation Engine

- ReportLab PDF Report (GET `/api/v1/reports/{session_id}/pdf`): Generates multi-page PDF featuring executive summary, severity metrics pie chart, findings table, remediation playbooks, and compliance gap analysis.
- Markdown Report (GET `/api/v1/reports/{session_id}/markdown`): Formatted briefing suitable for ticketing platforms (Jira, GitHub Issues).
- JSON Report (GET `/api/v1/reports/{session_id}/json`): Serialized SentinelState Pydantic v2 data model.

3. Session Risk Rating Matrix

- CRITICAL Risk: Assigned if ≥ 1 Critical finding or active breach detected.
- HIGH Risk: Assigned if ≥ 1 High severity vulnerability or unpatched CVE exists.
- MEDIUM / LOW Risk: Assigned for non-critical hardening recommendations.